# A proof of the Caporossi–Hansen conjecture on the Randić index and graph radius

Keio Shaitetsu

13 November 2026 · Research manuscript for independent review

**Abstract**

For every finite connected simple graph of maximum degree $\Delta \geq 2$, we prove $R(G) - \mathrm{rad}(G) \geq \sqrt{\Delta} - 3/2$, where $R$ is the classical Randić index. Thus the paths of even order at least four are exactly the connected graphs whose Randić index is smaller than their vertex radius. Equality between the index and vertex radius holds exactly for the singleton, the single edge and the even cycles. The main estimate bounds the absolute radius $\rho(G)$ by the sum $M(G)$ of reciprocal maximum endpoint degrees. In a minimal counterexample, deleting the leaves attached to vertices below a degree threshold gives an upper bound for the number of edges crossing that threshold. Summing these bounds and constructing a spanning tree of the two-connected block gives $2\rho(G) \leq n - \Delta + 4 \leq 2M(G)$ in the proposed counterexample, where $n$ is the order of that block.

## 1 Introduction and principal results

For a finite connected undirected simple graph $G$, let $d_v$ denote the degree of a vertex $v$. Its Randić index and its vertex radius are

$$R(G) = \sum_{uv \in E(G)} \frac{1}{\sqrt{d_u d_v}}, \qquad \mathrm{rad}(G) = \min_{v \in V(G)} \max_{w \in V(G)} \mathrm{dist}_G(v, w).$$

Here $uv$ denotes an unordered edge, counted once. The one-vertex graph has radius zero and an empty edge sum. The notation $P_N$ and $C_N$ denotes a path and a cycle on $N$ vertices, respectively. Throughout, $\Delta$ denotes maximum degree and diam denotes diameter. The usual bound $\mathrm{diam}(G) \leq 2\,\mathrm{rad}(G)$ follows by joining two vertices through a vertex center.

Fajtlowicz proposed $R(G) \geq \mathrm{rad}(G) - 1$ [7]. Caporossi and Hansen conjectured $R(G) \geq \mathrm{rad}(G)$ for every connected graph other than a path of even order and proved the tree case [1, 9]. We obtain a bound in terms of maximum degree, as well as a complete classification.

**Theorem 1.1.** *Let $G$ be a finite connected simple graph with maximum degree $\Delta \geq 2$. Then*

$$R(G) - \mathrm{rad}(G) \;\geq\; \sqrt{\Delta} - \tfrac{3}{2}. \tag{1}$$

*For the class of all finite connected simple graphs, the following alternatives hold.*

1. *$R(G) < \mathrm{rad}(G)$ precisely for $P_{2m}$ with $m \geq 2$.*
2. *$R(G) = \mathrm{rad}(G)$ precisely for $K_1$, $K_2$, and $C_{2m}$ with $m \geq 2$.*
3. *Every other graph satisfies $R(G) > \mathrm{rad}(G)$.*

*The minimum of $R(G) - \mathrm{rad}(G)$ over all such graphs is $\sqrt{2} - 3/2$, attained precisely in the first case.*

The modified Randić index

$$M(G) = \sum_{uv \in E(G)} \frac{1}{\max(d_u, d_v)}$$

occurs in the diameter analysis of Dvořák, Lidický and Škrekovski [6] and in the radius argument of Cygan, Pilipczuk and Škrekovski [3]. The latter proved the Caporossi–Hansen inequality for graphs of maximum degree at most four and established $M(G) \geq \mathrm{rad}(G) - 1/2$ in that class. Zuo proved the conjecture for unicyclic and bicyclic graphs [9]. Deng, Tang and Zhang gave a bound depending on the cyclomatic number [4]. Doig studied the inequality for cactus graphs [5]. The spanning-tree estimate used in Section 2 also appears as Lemma 2.1 of [3].

Give every edge of $G$ length one and denote the resulting metric graph by $|G|$. Write

$$\rho(G) = \min_{x \in |G|} \max_{w \in V(G)} \mathrm{dist}_{|G|}(x, w).$$

Thus a center may lie inside an edge, and the vertices are the demand points.

**Theorem 1.2** (Absolute-radius inequality)**.** *Every finite connected simple graph satisfies* $M(G) \geq \rho(G)$.

The proof of Theorem 1.2 builds on the minimal-counterexample arguments in VMRC [087] and [106] and the local degree identity in VMRC [094] [8]. All reductions are proved in Sections 2–5. The core with attached leaves is a bag in the sense of Definition 4.1 of [3]. The step that removes the degree bound is Proposition 4.2, which compares each cut with the effect of deleting just the leaves on its lower-degree side. Section 6 then deduces Theorem 1.1 by a comparison edge by edge.

## 2 Metric centers and spanning trees

Distances between vertices in $|G|$ agree with graph distances. With finitely many vertices, the maximum of their distance functions is a continuous function on the compact space $|G|$, so an absolute center exists.

**Lemma 2.1.** *An absolute center can be chosen at a vertex or an edge midpoint. In particular* $\mathrm{rad}(G) - 1/2 \leq \rho(G) \leq \mathrm{rad}(G)$.

*Proof.* Consider an edge $uv$ with coordinate $t \in [0, 1]$ measured from $u$. For each vertex $w$ the distance from $t$ to $w$ is

$$\min\{\mathrm{dist}_G(u, w) + t,\ \mathrm{dist}_G(v, w) + 1 - t\}.$$

Both affine functions have integer intercepts and slopes $\pm 1$. Two such functions of opposite slopes can meet only at a half-integer. Consequently every individual distance, and their maximum, is affine on each of the open half-edges. Its minimum on the closed edge is attained at an endpoint or midpoint. Distances from a center so chosen to vertices are multiples of $1/2$. Rounding any absolute center to a nearest vertex increases all its distances by at most $1/2$. The vertex centers are allowed when minimizing for $\rho$. □

**Lemma 2.2.** *For a tree $T$ of diameter $L$, one has $\rho(T) = L/2$ and $\mathrm{rad}(T) = \lceil L/2 \rceil$. If a connected graph $H$ on $m$ vertices has a vertex with at least $j \geq 2$ incident edges, then it admits a spanning tree of diameter at most $m - j + 1$. In particular $2\,\mathrm{rad}(H) \leq m - j + 2$.*

*Proof.* Let a longest path in $T$ have endpoints $a, b$. For any vertex $w$, its branch meets the path at a unique point at distance $s$ from $a$; write $t$ for the distance from that point to $w$. The two paths to the endpoints give $t \le \min(s, L-s)$. Thus the distance from the midpoint to $w$ is at most $|s - L/2| + \min(s, L-s) = L/2$. The endpoints force a radius of at least $L/2$ by the triangle inequality. Rounding the midpoint yields the asserted integer radius.

The $j$ chosen incident edges form a tree. Extend this tree, one edge to a new vertex at a time, to a spanning tree $T$ of $H$. At least $j$ components of $T - v$ meet the chosen vertex $v$ by distinct edges. Each contains a leaf of $T$, as a farthest vertex of that component from $v$ is a leaf. A longest path in $T$ uses at most two leaves. It therefore uses at most $m - (j-2)$ vertices and has at most $m - j + 1$ edges. Distances in $H$ can only be shorter than those in its spanning tree. The vertex-radius bound now follows from $2\lceil L/2 \rceil \le L + 1$. □

## 3 Reductions of a minimal counterexample

Suppose Theorem 1.2 fails. Among counterexamples $G$ choose one minimizing the pair $(\beta(G), |E(G)|)$ lexicographically, where $\beta(G) = |E(G)| - |V(G)| + 1$ is the cyclomatic number. This order on pairs of nonnegative integers is well founded. A graph smaller in this order satisfies $M \ge \rho$. Stars with at least two leaves have $M = \rho = 1$. On a path of order $m \ge 3$, each edge contributes $1/2$ to $M$ and Lemma 2.2 gives $M = \rho = (m-1)/2$. Also $(M, \rho) = (0, 0)$ for $K_1$ and $(1, 1/2)$ for $K_2$.

**Lemma 3.1.** *If $e$ is a nonbridge edge of $G$, then $M(G-e) > M(G)$. If $z$ is a leaf with neighbor $v$ of degree $d \ge 2$, and $b$ of the remaining neighbors of $v$ have degree less than $d$, then*

$$M(G) - M(G-z) = \frac{1}{d} - \frac{b}{d(d-1)} \ge 0, \qquad 0 \le \rho(G) - \rho(G-z) \le \tfrac{1}{2}. \tag{2}$$

*The leaf formulas apply to any connected graph in the stated range.*

*Proof.* Deleting a nonbridge preserves the demand vertices, removes potential center locations, and cannot shorten distances. It reduces $\beta$ by one. Hence $M(G-e) \ge \rho(G-e) \ge \rho(G) > M(G)$.

Deleting $z$ removes $1/d$ from $M$. If the other endpoint of an edge at $v$ has degree less than $d$, its weight increases from $1/d$ to $1/(d-1)$. The other weights stay fixed. As $b \le d-1$, this proves the index formula.

Project a center lying on the removed pendant interval to $v$. All distances to retained vertices weakly decrease, proving the first radius inequality. Conversely, choose a vertex or midpoint center $y$ of $G - z$, and put $r_0 = \rho(G-z)$. If $\operatorname{dist}(y, v) \le r_0 - 1/2$, then $\operatorname{dist}(y, z) \le r_0 + 1/2$ after restoration. Otherwise $\operatorname{dist}(y, v) = r_0$ by Lemma 2.1. As $G - z$ has at least two vertices, $r_0 \ge 1/2$. Move $y$ a distance $1/2$ along a shortest path toward $v$. The distances to retained vertices increase by at most $1/2$, and the distance to $z$ is $r_0 + 1/2$. □

A vertex adjacent to a leaf will be called its support. A support of degree two in $G$ has a second neighbor of degree at least two, because $G$ is not a star. Formula (2) then gives a drop in $M$ of $1/2$, at least the drop in $\rho$. Deleting this leaf would produce a smaller counterexample with the same $\beta$. Thus every support in $G$ has degree at least three.

If $z, z'$ are leaves with the same support $v$, delete $z$. In the remaining graph a center can be chosen outside the pendant interval leading to $z'$, except possibly at $v$. Indeed, projecting a point of that interval to $v$ decreases distances to all other vertices, and puts $z'$ at distance one. Any point on that interval has eccentricity at least one, since another neighbor of $v$ remains. Restoring $z$ now gives $z$ the same distance from the center as $z'$. The radius is unchanged and $M$ weakly decreases upon deletion. Minimality gives a contradiction. Each support therefore has exactly one leaf.

**Lemma 3.2** (Splitting at a cut vertex)**.** *For every vertex $v$ of the minimal counterexample $G$, at most one component of $G - v$ contains an edge.*

*Proof.* Otherwise partition $V(G) \setminus \{v\}$ into unions $A, B$ of components of $G - v$, each containing an edge. For $i \in \{A, B\}$ let $t_i$ be the number of neighbors of $v$ in $i$, and $m_i$ the number of edges within $i$. All four numbers are positive. Form $G_A$ by retaining $G[A \cup \{v\}]$ and adding $t_B$ new leaves at $v$. Define $G_B$ symmetrically. These graphs are connected, $|E(G_A)| = |E(G)| - m_B$, and $\beta(G_A) = m_A + t_A - |A|$. The analogous expression for $G_B$ gives $\beta(G_A) + \beta(G_B) = \beta(G)$. Both graphs are therefore smaller. Each retained edge has its original endpoint degrees. The $t_A + t_B = d_v$ artificial edges across the two graphs each have weight $1/d_v$, which gives

$$M(G_A) + M(G_B) = M(G) + 1. \tag{3}$$

Set $r_i = \rho(G_i)$. In each $G_i$, an artificial leaf and a retained neighbor of $v$ are at distance two, so $r_i \geq 1$. Choose a center $y_i$ in the retained part: if a center lies on an artificial pendant edge, project it to $v$. This weakly decreases its distances to all vertices other than the tip of that edge, and places that tip at distance $1 \leq r_i$. If $s_i = \operatorname{dist}(y_i, v)$, the distance to an artificial leaf is $s_i + 1$, so $s_i \leq r_i - 1$.

In $|G|$ join $y_A$ to $v$ and $v$ to $y_B$ by shortest paths. Their concatenation is a geodesic since the two retained parts meet only at $v$. With signed distance $u$ from $v$ along this path, its ends have coordinates $-s_A, s_B$. Via $y_A$ and $y_B$, respectively, the distances to the original vertices on the two sides (including $v$) are at most

$$L(u) = r_A + s_A + u, \qquad H(u) = r_B + s_B - u.$$

If these functions meet on the interval, their common value is $(r_A + s_A + r_B + s_B)/2 \leq r_A + r_B - 1$. If $L$ dominates throughout, at $u = -s_A$ their maximum is $r_A$. If $H$ dominates throughout, at $u = s_B$ the maximum is $r_B$. Both are at most $r_A + r_B - 1$. Consequently $\rho(G) \leq r_A + r_B - 1 \leq M(G_A) + M(G_B) - 1 = M(G)$ by minimality and (3), a contradiction. □

**Proposition 3.3.** *The minimal counterexample consists of a two-connected induced subgraph $K$ of order at least three together with at most one leaf attached to each vertex of $K$. Every edge of $K$ is a nonbridge in $G$.*

*Proof.* A tree containing a path of length four violates Lemma 3.2 at the middle of that path. A tree of diameter at most two is a singleton, an edge or a star. In a tree of diameter three, every vertex lies on the central edge of a diametral path or is adjacent to one of its ends: the midpoint covers the tree within 3/2 by Lemma 2.2. Each central vertex supports at most one leaf, so this tree is a path. These possibilities were excluded above.

Thus $G$ contains a cycle. Its vertices induce a two-connected graph, where two-connected means having at least three vertices and remaining connected after deletion of any one vertex. Extend that set to a maximal vertex set inducing a two-connected subgraph $K$. If a component outside $K$ has two distinct attachment vertices in $K$, it gives a path between them with all interior vertices outside $K$. Adding the path preserves two-connectivity: deletion of an interior vertex leaves each remaining piece attached at an end, and deletion of a vertex of $K$ leaves $K$ connected with at least one end of the path attached. This contradicts maximality. Each outside component therefore has a unique attachment $v$. If it contained an edge, deleting $v$ would separate that edge from $K - v$, which is connected and has at least two vertices. Lemma 3.2 excludes this. Each outside component is a single leaf, and the previous reduction bounds their number per support. Finally, every vertex of $K$ has at least two neighbors in $K$. For a core edge $uv$, choose another core neighbor $w$ of $u$; a path in $K - u$ from $w$ to $v$ completes an alternative route from $u$ to $v$. Hence $uv$ is a nonbridge. □

# 4 A bound for each degree cut

Fix the subgraph $K$ of Proposition 3.3 and write $n = |V(K)|$. Degrees $d_v$ in this section are always measured in $G$. At a core vertex $v$, let $p_v$ be 1 if $v$ supports a leaf and 0 otherwise. Define

$$\ell_v = \#\{w \sim v : d_w < d_v\}, \quad h_v = d_v - \ell_v, \quad g_v = d_v(d_v - 1), \quad a_v = \ell_v/g_v, \quad b_v = h_v/g_v.$$

In particular $\ell_v$ includes the leaf when $p_v = 1$.

The next lemma includes the local-minimum and minimum-degree properties of bags from Lemma 4.2 of [3].

**Lemma 4.1.** *Every core vertex has $d_v \geq 3$ and $\ell_v \geq 1$. For a core edge $uv$ with $d_u < d_v$ we have $a_u > b_v$.*

*Proof.* On deleting $uv$, each remaining edge at an endpoint $x$ gains $1/g_x$ precisely when its other endpoint has degree strictly below $d_x$. No remaining edge occurs at both $u$ and $v$, since $G$ is simple. Subtracting the weight of the deleted edge gives

$$M(G - uv) - M(G) = a_u + \frac{\ell_v - 1}{g_v} - \frac{1}{d_v} = a_u - b_v \qquad (d_u < d_v), \tag{4}$$

$$= \frac{\ell_u + \ell_v}{d(d-1)} - \frac{1}{d} \qquad (d_u = d_v = d). \tag{5}$$

Both changes must be positive by Lemma 3.1. If $\ell_u = 0$, there is no leaf at $u$. A core neighbor of higher degree would contradict (4), since $b_v \geq 0$. A neighbor of equal degree would have $\ell_v \leq d - 1$, contradicting (5). These exhaust the neighbors of $u$. Finally, a vertex of full degree two already has two neighbors in $K$; all vertices of $K$ have degree at least two. It could have no lower-degree neighbor. □

A core vertex of minimum full degree must have a leaf, by Lemma 4.1. Thus $K$ is smaller than $G$ in the order of Section 3: removing the leaves leaves $\beta$ unchanged and reduces the edge count. Let $\delta = M(G) - M(K)$. Repeated leaf deletion and minimality give

$$0 \leq \delta < \rho(G) - \rho(K) \leq 1. \tag{6}$$

The last bound holds because a center of $K$ reaches each added leaf within $\rho(K) + 1$. If $\Delta = 3$, every core vertex has full degree three. Its only possible lower-degree neighbor is its leaf, giving $\ell_v = 1$ at every core vertex. Equation (5) gives a zero change at every core edge, a contradiction. Hence $\Delta \geq 4$.

**Proposition 4.2** (Degree-cut estimate)**.** *For an integer $3 \leq k < \Delta$ let $S_k = \{v \in V(K) : d_v \leq k\}$, and let $N_k$ count edges of $K$ with exactly one end in $S_k$. Then*

$$N_k \leq k(k-1)\delta. \tag{7}$$

*Proof.* Write $S = S_k$. Orient each unequal-degree core edge from its lower endpoint to its higher endpoint. Denote by $I$ the directed edges inside $S$ and by $F$ the directed edges leaving $S$. Every crossing edge leaves $S$. Let $e_v$ count the core neighbors with degree $d_v$. At $v$ there are $\ell_v - p_v$ incoming and $h_v - e_v$ outgoing edges. Since $h_v a_v = \ell_v b_v$, we have

$$p_v b_v = (h_v - e_v)a_v - (\ell_v - p_v)b_v + e_v a_v.$$

Sum this identity over $v \in S$. An internal directed edge $u \to v$ contributes $a_u - b_v$, and an edge leaving $S$ contributes $a_u$:

$$\sum_{v \in S} p_v b_v = \sum_{(u,v) \in I} (a_u - b_v) + \sum_{(u,v) \in F} a_u + \sum_{v \in S} e_v a_v. \tag{8}$$

Each term of the first sum is positive by Lemma 4.1.

Delete precisely the leaves whose support belongs to $S$ and call the result $G_S$. To compute the change of $M$, first consider a deleted leaf at $v$. It has weight $1/d_v$, and the $\ell_v - 1$ edges to strictly lower core neighbors each gain $1/g_v$. Indeed, on a strict edge with original degrees $d < e$, the new maximum remains $e$ when the higher endpoint keeps its leaf or has none. If the higher endpoint loses a leaf, the new maximum is $e-1$, as $d \le e-1$, irrespective of any deletion at the lower endpoint. Thus the loss from that leaf, after these gains, is $1/d_v - (\ell_v - 1)/g_v = b_v$.

On a core edge of equal degrees $d$, the maximum decreases precisely when both endpoints lose a leaf. In that case the edge gains $1/[d(d-1)]$. Let $Q$ consist of these unordered core edges. The exact total change is therefore

$$M(G) - M(G_S) = \sum_{v \in S} p_v b_v - \sum_{uv \in Q} \frac{1}{g_u}. \tag{9}$$

For $uv \in Q$, the last sum of (8) counts $a_u + a_v$ on that edge. This is at least $2/g_u$ since $g_u = g_v$ and $\ell_u, \ell_v \ge 1$. All other contributions to that last sum are nonnegative. Subtracting the correction in (9) from (8), and then deleting the remaining leaves one at a time by Lemma 3.1, gives

$$\sum_{(u,v) \in F} a_u \le M(G) - M(G_S) \le M(G) - M(K) = \delta.$$

For every $(u, v) \in F$, one has $\ell_u \ge 1$ and $d_u \le k$, so that $a_u \ge 1/[k(k-1)]$. There are $N_k$ such edges, giving (7). □

## 5 Summing the cuts and bounding the absolute radius

*Completion of the proof of Theorem 1.2.* Continue with the hypothetical minimal counterexample. Write $t = \sum_{v \in V(K)} p_v$, so $G$ has $n + t$ vertices. In a graph without isolated vertices, the sum of $1/d_u + 1/d_v$ over the edges counts each vertex once. Subtracting $2M(G)$ gives

$$n + t - 2M(G) = \sum_{uv \in E(G)} \left| \frac{1}{d_u} - \frac{1}{d_v} \right|.$$

A leaf at $v$ contributes $1 - 1/d_v$. A core edge with degrees $d < e$ contributes $\sum_{k=d}^{e-1} 1/[k(k+1)]$, counting exactly the cuts it crosses. Equal-degree edges contribute zero. We obtain

$$n - 2M(G) = \sum_{k=3}^{\Delta-1} \frac{N_k}{k(k+1)} - \sum_{v \in V(K)} \frac{p_v}{d_v}. \tag{10}$$

When all leaves are deleted, the retained core-edge weights can only increase. Thus the sum of the removed weights $\sum_v p_v/d_v$ is at least $\delta$. By Proposition 4.2,

$$n - 2M(G) \le \left( \sum_{k=3}^{\Delta-1} \frac{k-1}{k+1} - 1 \right) \delta \le (\Delta - 4)\delta \le \Delta - 4. \tag{11}$$

There are $\Delta - 3$ summands, each at most one; the last two inequalities use $\delta \ge 0$, $\Delta \ge 4$ and (6).

Choose a core vertex of full degree $\Delta$. It has at least $\Delta - 1$ neighbors in $K$, since at most one is a leaf. By Lemma 2.2, $K$ has a spanning tree of diameter at most $n - (\Delta - 1) + 1 = n - \Delta + 2$. Restore each original leaf and its incident edge to that tree. Between any two vertices of the resulting tree $T$, the path consists of a core-tree path and at most two leaf edges. Thus $\operatorname{diam}(T) \le n - \Delta + 4$. Distances in $G$ from a diametral midpoint of $T$ are at most their distances in $T$. Applying Lemma 2.2 and (11) gives

$$2\rho(G) \le \operatorname{diam}(T) \le n - \Delta + 4 \le 2M(G). \tag{12}$$

This contradicts $M(G) < \rho(G)$ and proves Theorem 1.2. □

## 6 From absolute radius to vertex radius

Let $N = |V(G)|$, $r = \mathrm{rad}(G)$, and assume $\Delta \geq 2$. We first keep the difference $r - \rho(G)$ in the estimate; on graphs with a vertex absolute center this term vanishes.

**Proposition 6.1.** *For each such graph,*

$$R(G) - r \;\geq\; \frac{N - 2r - (\sqrt{\Delta} - 1)(r - \rho(G))}{\sqrt{\Delta} + 1}. \tag{13}$$

*Proof.* The degree-square identity of Caporossi, Gutman, Hansen and Pavlović [2] expresses $N - 2R(G)$ as a sum of squares. For clarity we derive it here and compare the summands, following the bounded-degree argument in Section 6 of [3]. On an edge with degrees $d \leq e$, the contribution to $R(G) - M(G)$ is $1/\sqrt{de} - 1/e \geq 0$, and factoring gives

$$(d^{-1/2} - e^{-1/2})^2 = (\sqrt{e/d} - 1)\left(\frac{1}{\sqrt{de}} - \frac{1}{e}\right).$$

Summing the expanded squares counts each vertex once in the terms $1/d_u + 1/d_v$. Since $\sqrt{e/d} \leq \sqrt{\Delta}$, we have

$$N - 2R(G) = \sum_{uv \in E(G)} (d_u^{-1/2} - d_v^{-1/2})^2 \leq (\sqrt{\Delta} - 1)(R(G) - M(G)).$$

Put $s = R(G) - r$ and $\eta = r - \rho(G)$. Theorem 1.2 gives $R(G) - M(G) \leq s + \eta$, whence $N - 2r - 2s \leq (\sqrt{\Delta} - 1)(s + \eta)$. Solving for $s$ gives (13). □

*Proof of Theorem 1.1.* Apply Lemma 2.2 with $j = \Delta$ to obtain $N - 2r \geq \Delta - 2$. Lemma 2.1 gives $0 \leq r - \rho(G) \leq 1/2$. Substitution in (13) yields

$$R(G) - r \geq \frac{\Delta - 2 - (\sqrt{\Delta} - 1)/2}{\sqrt{\Delta} + 1} = \sqrt{\Delta} - \tfrac{3}{2}.$$

This is strictly positive for $\Delta \geq 3$. A connected graph of maximum degree two is a path or a cycle: following successive edges either closes a cycle, which leaves no available edge to another vertex, or ends at the two ends of a spanning path. For $N \geq 3$ a path has two edges of degrees $(1, 2)$ and $N - 3$ of degrees $(2, 2)$. In a cycle all edge degrees are $(2, 2)$. Their vertex radii equal $\lfloor N/2 \rfloor$, by Lemma 2.2 on a path and by measuring the two arc lengths on a cycle. The resulting values are

| | $N$ even | $N$ odd |
|---|---|---|
| $R(P_N) - \mathrm{rad}(P_N)$ | $\sqrt{2} - 3/2$ | $\sqrt{2} - 1$ |
| $R(C_N) - \mathrm{rad}(C_N)$ | $0$ | $1/2$ |

The singleton has $R = r = 0$ and the single edge has $R = r = 1$. These cases exhaust all connected simple graphs. Since $\sqrt{2} - 3/2 < 0$, the first entry gives exactly the exceptional paths and the global minimum. □

The coefficient of $\sqrt{\Delta}$ in (1) cannot be increased while retaining a constant additive term. For $\Delta \geq 3$, subdivide one edge of the star $K_{1,\Delta}$ once. Its diameter is three and its vertex radius is two. There are $\Delta - 1$ edges of degrees $(1, \Delta)$, one of degrees $(2, \Delta)$ and one of degrees $(1, 2)$. For this family,

$$R(G) - r = \sqrt{\Delta} - 2 + \frac{1}{\sqrt{2}} - \frac{1 - 1/\sqrt{2}}{\sqrt{\Delta}}.$$

## References

[1] G. Caporossi and P. Hansen. Variable neighborhood search for extremal graphs 1: The AutoGraphiX system. *Discrete Mathematics* **212** (2000), 29–44. doi:10.1016/S0012-365X(99)00206-X.

[2] G. Caporossi, I. Gutman, P. Hansen and L. Pavlović. Graphs with maximum connectivity index. *Computational Biology and Chemistry* **27** (2003), 85–90. doi:10.1016/S0097-8485(02)00016-5.

[3] M. Cygan, M. Pilipczuk and R. Škrekovski. On the inequality between radius and Randić index for graphs. *MATCH Communications in Mathematical and in Computer Chemistry* **67** (2012), 451–466. Journal PDF.

[4] H. Deng, Z. Tang and J. Zhang. On a conjecture of Randić index and graph radius. *Filomat* **29** (2015), 1369–1375. doi:10.2298/FIL1506369D.

[5] M. I. Doig. Block eccentricity, a radius bound, and an application to the Randić index. Preprint (2023). arXiv:2309.11613.

[6] Z. Dvořák, B. Lidický and R. Škrekovski. Randić index and the diameter of a graph. *European Journal of Combinatorics* **32** (2011), 434–442. doi:10.1016/j.ejc.2010.12.002.

[7] S. Fajtlowicz. On conjectures of Graffiti. *Discrete Mathematics* **72** (1988), 113–118. doi:10.1016/0012-365X(88)90199-9.

[8] Keio. *Verifiable Mathematical Reasoning Corpus*. Research drafts [087], [094] and [106]. Repository snapshot 8a3f6895d0b9.

[9] L. Zuo. About a conjecture on the Randić index of graphs. *Bulletin of the Malaysian Mathematical Sciences Society* (2) **35** (2012), 411–424. Journal PDF.